Fiscal Year:	FY 2010	**Task Last Updated:**	FY 10/08/2009
PI Name:	Strangman, Gary E Ph.D.		
Project Title:	Objective Detection, Evaluation and Countermeasures for In-flight Depression		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	strang@nmr.mgh.harvard.edu	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	617-724-0662
Organization Name:	Massachusetts General Hospital		
PI Address 1:	Department of Psychiatry		
PI Address 2:	149 13th Street, Suite 2651		
PI Web Page:			
City:	Charlestown	**State:**	MA
Zip Code:	02129-2020	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 NSBRI-RFA-07-01 Human Health in Space
Start Date:	10/01/2007	**End Date:**	09/30/2011
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Cartreine (nee Carter), James (Beth Israel Deaconess Medical Center) Yeung, Albert (Massachusetts General Hospital) Fava, Maurizio (Massachusetts General Hospital) Feldman, Gregory (Massachusetts General Hospital) Zeffiro, Thomas (Argosy Omnimedia, Inc.) Zhang, Quan (Massachusetts General Hospital)		
Grant/Contract No.:	NCC 9-58-NBPF01301		
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Task Description:	Spaceflight stressors, including depression, can significantly disrupt one's ability to function effectively and efficiently, and associated performance deficits can seriously jeopardize space mission success. Mission success can be jeopardized either directly, from the potentially life threatening consequences of lapses in performance, or indirectly, by adding to the workload and stress of other crewmembers. The substantial likelihood and potentially serious consequences of depression during spaceflight explains why Bioastronautics Roadmap Risk #25--namely, human performance failure due to neurobehavioral problems--is a high priority risk for all mission types (ISS, Moon, Mars). A variety of therapies are already available, including preventative measures, medications and psychological consultations with ground-crews. However, current methods to decide whether a therapy needs to be used rely heavily on subjective self-report. The biological basis of mood disorders suggests neural biomarkers may provide a more objective method for assessing depression and associated performance deficits. This proposal aims to identify neural biomarkers sensitive to, and specific for both depression detection and depression severity assessment. Since there is currently no reliable way to monitor brain status or function in spaceflight, an important component of our project will be technology development focused on the evaluation and validation of a flight-capable, noninvasive neuroimaging technology: near-infrared neuroimaging (NIN). The aims for this project therefore seek to evaluate the ability of near-infrared neuroimaging to more objectively detect clinical depression and assess its severity, and to improve our existing mobile near-infrared neuroimaging technology to enable more robust mobile monitoring of brain hemodynamics. In year one, we focused on development activities to support a ground-based neuroimaging study of depression. This included development and testing of sensitive tasks for brain function assessment, optimizing neuroimaging protocols, and creating integrated stimulus display and data acquisition systems to enable continuous monitoring of complex bimanual inputs from our study participants during task performance. We also integrated with Dr. James Cartriene's clinical trial of his computer based problem solving therapy for depression. These efforts were designed to generate a rich dataset to quantify the performance effects of a stressor (depression) on neural activity and during complex cognitive-motor performance tasks. In year two, we began our neuroimaging data collection, and also focused on NIN technology development. We completed a second generation NIN prototype device and successfully tested it both in the laboratory and during a parabolic flight for sensitivity to brain hemodynamics. In the coming year, we plan to complete the neuroimaging of depressed participants, and will continue to advance the capabilities for NIN in mobile and spaceflight environments. If successful, these activities are expected to have two primary impacts. First, identification of brain-based biomarkers for depression will move the countermeasure readiness level (CRL) of brain-based evaluation of depression from CRL 2 to CRL 3. Once depression is objectively identified, there are already suitable depression countermeasures available for deployment in flight (including medications and psychological consults), plus Dr. Cartriene's computer-based therapy in development. Identification of brain based biomarkers would be useful not only in spaceflight, but for the millions of individuals suffering from depression on Earth, by providing a more objective and potentially more readily accessible method for evaluating depression. Second, our technology development efforts have already moved the technology readiness level (TRL) of near-infrared neuroimaging to TRL 5 and we are expecting to reach TRL 6 with coming advances and studies. Our use of relatively inexpensive and unobtrusive NIN technology will enable brain imaging and monitoring not only during spaceflight, but also in a variety of Earth-based contexts including in-office neuroimaging, rural areas, and even under-served communities or first-responder contexts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Successful completion of this project would lead to the following Earth-relevant benefits: 1. Depression Biomarkers: We seek to identify biomarkers that are suitable for (i) depression diagnosis, and (ii) assessing depression severity. If a reliable NIN-based biomarker is identified, this would provide initial validation of the lower cost NIN-based evaluation of depressed individuals. Such brain-based biomarkers could then be measured in an office setting and at relatively low cost, thereby enabling access to such capabilities in broader regions than currently possible, including rural or underserved communities. 2. Mobile Neuroimaging: Developing appropriate technologies can enable neuroimaging in mobile environments, including spaceflight analogs and spaceflight itself. Such technologies have the potential to impact a wide range of novel brain monitoring applications on Earth as well, ranging from mobile epilepsy monitoring, to monitoring treatment efficacy via brain imaging in a doctor's office, to battlefield or first-responder head trauma evaluations, as well as generally more available, less expensive methods for diagnosing, monitoring, and treating depression or other disorders involving alterations in brain function. NIN is of particular promise as a brain imaging technology as it is sufficiently low-cost, robust and portable to be made readily available in diverse operational environments including urban, rural, and remote settings.
Task Progress:	In the past year, we have made significant progress on three primary fronts: experimental, computational, and device development. On the experimental front we completed development of optimized neuroimaging protocols and tasks, and began neuroimaging data collection for the depression study. Initial results suggest that both our MRI and near-infrared neuroimaging (NIN) measures are quite sensitive to task-related brain activity alterations in depressed participants. The tasks we have implemented will allow us to examine complex sensorimotor performance in much finer detail than previously possible. We also completed data collection for a laboratory test of NIN under head-down tilt conditions. This study is intended to quantify any sensitivity changes in NIN measurements in the presence of the fluid shifts toward the head expected during spaceflight. On the computational front, we completed three major subtasks, including (1) developing software for advanced artifact filtering of NIN data (via an NSBRI-funded student project supporting Mr. Neil Parikh), (2) initial development of software for integrated near-infrared neuroimaging data analysis and display (and an associated peer-reviewed publication), and (3) 18,000 hours of computer processing time to generate models of the distribution of photons migrating through the head from each of 3,555 separate injection points (5mm spacing) around the entire scalp. The simulation data provide by far the most detailed maps of brain sensitivity available for NIN measurements. Together, these three accomplishments will enable significantly more accurate and more automated NIN data recording, analysis and display. On the device development front, we achieved two major milestones regarding lightweight, mobile recording devices. First, on December 23, 2008, we successfully collected data from the brain using our newest portable near-infrared neuroimaging (NIN) instrument, OpticHolter version 2a. This wearable device provides up to four optical and four electrophysiological recording channels (ECG, respiration and two accelerometer channels), all in a package that weighs under 350 grams. Second, on June 21, 2009, we were able to successfully test our OpticHolter 2a device for synchronous

	recording of NIN-based hemodynamics, ECG, respiration and dual-axis accelerometry during a ZeroG parabolic flight. Clear changes in cardiac activity, respiration and hemodynamic fluctuations measured from the head were recorded, and all of these parameters demonstrated changes associated with the gravitational transitions generated by the aircraft flight pattern. These efforts strongly support the goal of developing technologies suitable for brain imaging in spaceflight.
Bibliography Type:	Description: (Last Updated: 07/08/2026)
Abstracts for Journals and Proceedings	Strangman GE, Zhang Q, Zeffiro TA. "Cognitive and sensorimotor assessment via mobile functional near-infrared neuroimaging." 80th Annual Aerospace Medical Association Scientific Meeting, Los Angeles, CA, May 4-7, 2009. Aviat Space Environ Med. 2009 Mar;80(3):287. , Mar-2009
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Articles in Peer-reviewed Journals	Strangman GE, Zhang Q, Zeffiro T. "Near-infrared neuroimaging with NinPy." Frontiers in Neuroinformatics 2009;3:12. Published online 2009 May 29. http://dx.doi.org/10.3389/neuro.11.012.2009 ; PMCID: PMC2698776 , May-2009